

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC 95% CI	Samples	Positive Rate	Cross-Val Folds
Attention Pool Extra3	A Lines	Random Forest	0.758	[0.726-0.790]	43689	0.472	5
Attention Pool Extra3	A Lines	Logistic Regression	0.795	[0.778-0.813]	43689	0.472	5
Attention Pool Extra3	A Lines	XGBoost	0.794	[0.779-0.810]	43689	0.472	5
Attention Pool Extra3	Large Consolidation	Random Forest	0.863	[0.847-0.880]	43689	0.110	5
Attention Pool Extra3	Large Consolidation	Logistic Regression	0.919	[0.913-0.924]	43689	0.110	5
Attention Pool Extra3	Large Consolidation	XGBoost	0.901	[0.886-0.915]	43689	0.110	5
Attention Pool Extra3	Pleural Effusion	Random Forest	0.674	[0.602-0.745]	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	Logistic Regression	0.808	[0.759-0.856]	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	XGBoost	0.829	[0.778-0.880]	43689	0.011	5
Attention Pool Extra3	Other Pathology	Random Forest	0.644	[0.614-0.674]	43689	0.405	5
Attention Pool Extra3	Other Pathology	Logistic Regression	0.690	[0.662-0.717]	43689	0.405	5
Attention Pool Extra3	Other Pathology	XGBoost	0.684	[0.662-0.707]	43689	0.405	5
Original	A Lines	Random Forest	0.704	[0.660-0.749]	43689	0.472	5
Original	A Lines	Logistic Regression	0.761	[0.725-0.796]	43689	0.472	5
Original	A Lines	XGBoost	0.745	[0.702-0.788]	43689	0.472	5
Original	Large Consolidation	Random Forest	0.812	[0.760-0.865]	43689	0.110	5
Original	Large Consolidation	Logistic Regression	0.899	[0.877-0.922]	43689	0.110	5
Original	Large Consolidation	XGBoost	0.868	[0.856-0.881]	43689	0.110	5
Original	Pleural Effusion	Random Forest	0.663	[0.596-0.730]	43689	0.011	5
Original	Pleural Effusion	Logistic Regression	0.846	[0.817-0.876]	43689	0.011	5
Original	Pleural Effusion	XGBoost	0.834	[0.785-0.883]	43689	0.011	5
Original	Other Pathology	Random Forest	0.617	[0.604-0.630]	43689	0.405	5
Original	Other Pathology	Logistic Regression	0.647	[0.625-0.669]	43689	0.405	5
Original	Other Pathology	XGBoost	0.646	[0.635-0.658]	43689	0.405	5
Original Noinitweights	A Lines	Random Forest	0.702	[0.610-0.793]	43689	0.472	5
Original Noinitweights	A Lines	Logistic Regression	0.753	[0.688-0.817]	43689	0.472	5
Original Noinitweights	A Lines	XGBoost	0.737	[0.651-0.823]	43689	0.472	5
Original Noinitweights	Large Consolidation	Random Forest	0.747	[0.635-0.858]	43689	0.110	5
Original Noinitweights	Large Consolidation	Logistic Regression	0.867	[0.826-0.908]	43689	0.110	5
Original Noinitweights	Large Consolidation	XGBoost	0.806	[0.730-0.882]	43689	0.110	5
Original Noinitweights	Pleural Effusion	Random Forest	0.613	[0.552-0.674]	43689	0.011	5
Original Noinitweights	Pleural Effusion	Logistic Regression	0.843	[0.782-0.903]	43689	0.011	5
Original Noinitweights	Pleural Effusion	XGBoost	0.785	[0.728-0.842]	43689	0.011	5
Original Noinitweights	Other Pathology	Random Forest	0.601	[0.561-0.640]	43689	0.405	5
Original Noinitweights	Other Pathology	Logistic Regression	0.653	[0.609-0.697]	43689	0.405	5
Original Noinitweights	Other Pathology	XGBoost	0.640	[0.604-0.676]	43689	0.405	5